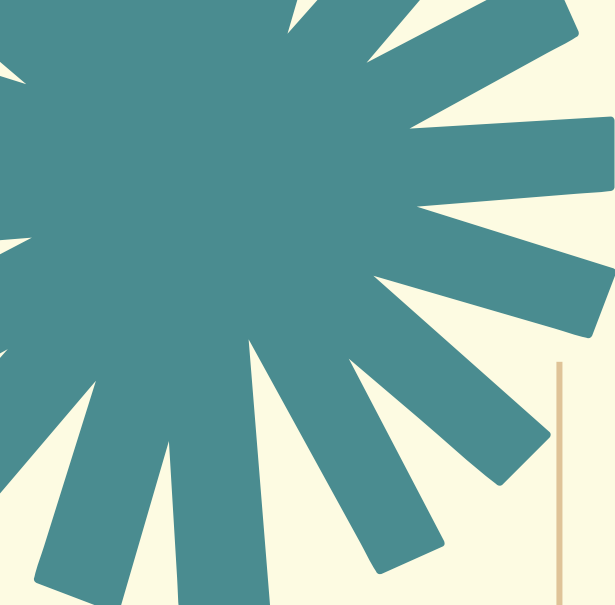
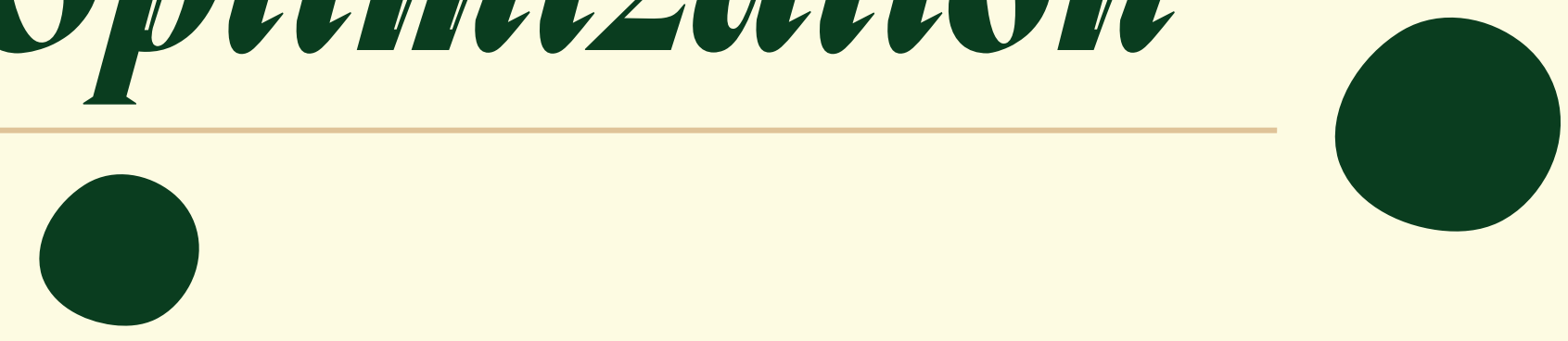
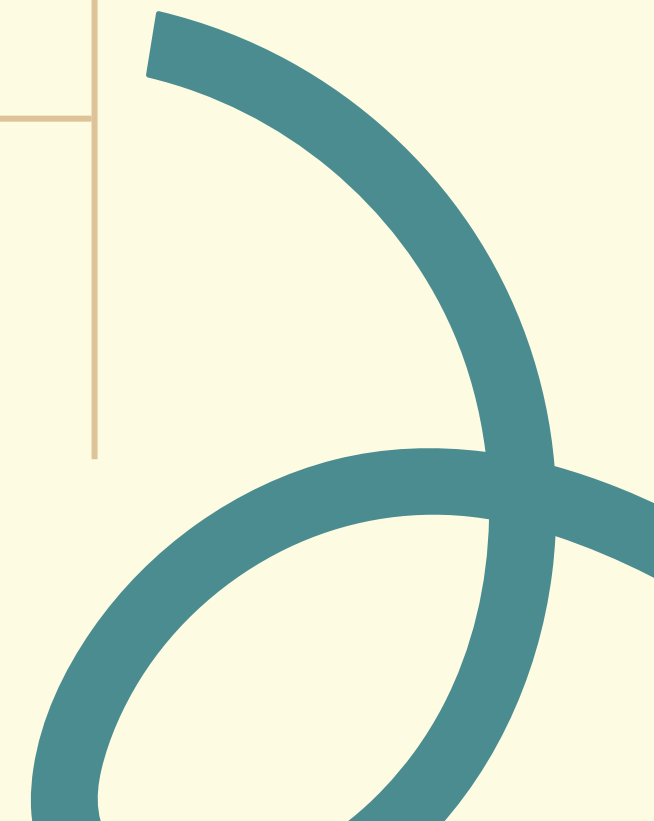




DATA ANALYTICS PORTFOLIO | RAFLI ALFIANSYAH
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Online Retail Strategy & Customer Behavior Optimization





Dataset Overview & Technical Stack

- **The Data:** Analyzed a comprehensive transactional dataset from a UK-based non-store online retail platform.
 - **Volume:** 541,191 rows of transaction data spanning from December 2010 to December 2011.
 - **The Data:** Analyzed a comprehensive transactional dataset from a UK-based non-store online retail platform.
 - **Market Reach:** Covering transactions across 38 countries, with a primary focus on the United Kingdom, Ireland, and Germany.
 - **Technical Ecosystem:**
 - **SQL (MySQL):** Used for intensive Data Cleaning (ETL) and building the RFM logic.
 - **Power BI:** Utilized for interactive storytelling and strategic dashboarding.
- 

The Challenge

Despite having a vast amount of data, the business lacked a clear understanding of its Customer Lifecycle. Marketing efforts were "one-size-fits-all," leading to high customer churn and inefficient inventory management.

- **KEY OBJECTIVES:**

- **Customer Intelligence:** Use RFM (Recency, Frequency, Monetary) modeling to segment the customer base.
- **Revenue Drivers:** Identify high-impact products and peak seasonality patterns to optimize stock.
- **Growth Strategy:** Develop tailored recommendations for "Champions" (retention) and "At Risk" (win-back) segments to maximize Customer Lifetime Value (CLV)

Data Transformation & Engineering Pipeline



Challenge

01 Missing Identifiers: Identified 135,080 rows (approx. 25% of data) with null CustomerID

02 Negative Values: Handled transactions with negative Quantity (Returns) and UnitPrice (Adjustments).

03 Inconsistent Formats: Converted string-based InvoiceDate into standardized SQL DateTime.

Solution

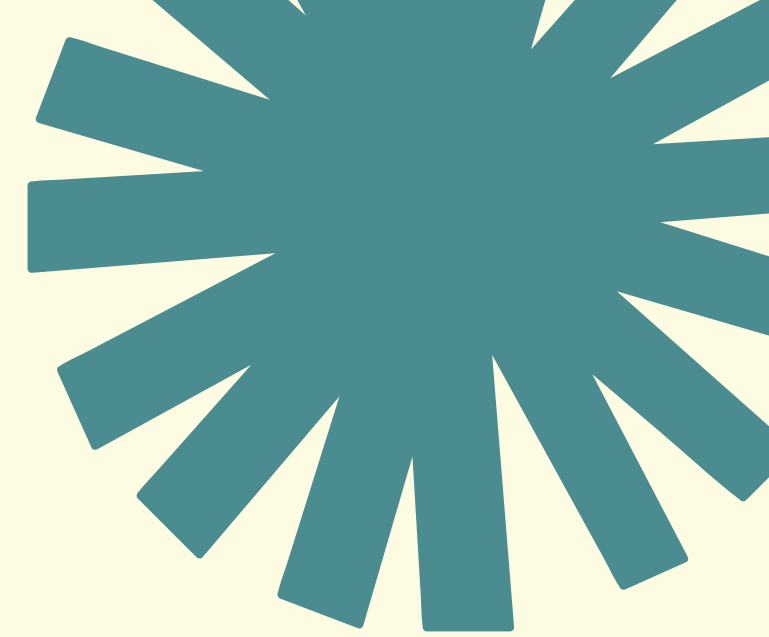
01 Sanitization: Filtered out null IDs for behavioral analysis to ensure segment accuracy.

02 Feature Engineering: Calculated Revenue at the database level (Quantity * UnitPrice) to optimize Power BI performance.

03 Dual-Track Modeling: Created a "Cleaned Master Table" for general sales and a "Registered Table" specifically for RFM & Cohort analysis.



SQL Query Syntax



```
-- Converting a string into the correct DATETIME format
UPDATE retail_data
SET InvoiceDate = STR_TO_DATE(InvoiceDate, '%d/%m/%Y %H:%i')
WHERE InvoiceDate LIKE '%/%';

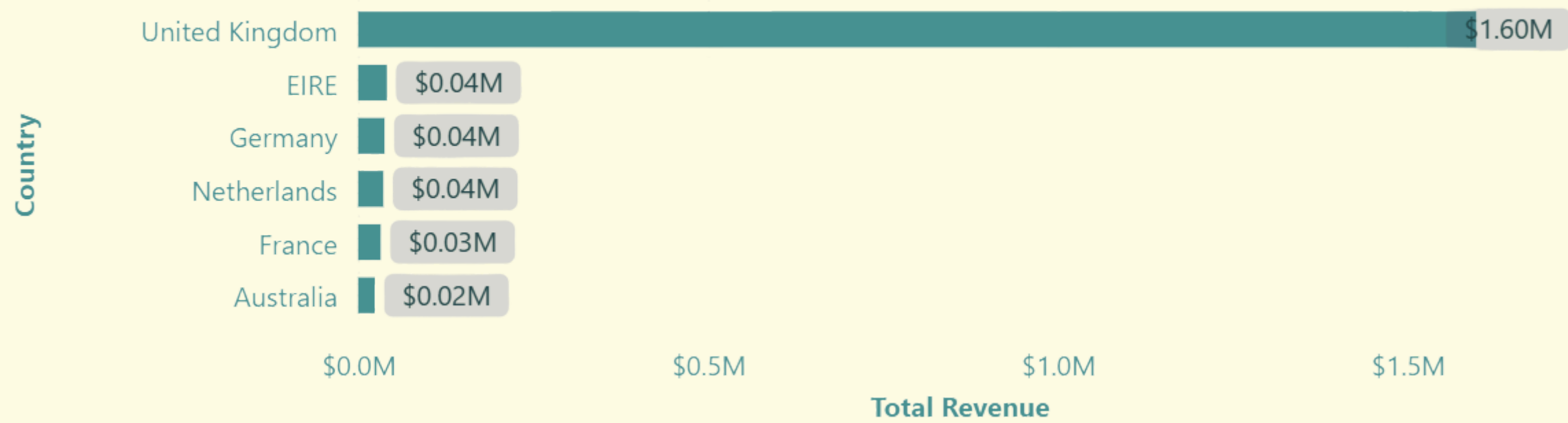
ALTER TABLE retail_data MODIFY COLUMN InvoiceDate DATETIME;
```



```
-- Eliminate failed transactions & handle missing CustomerIDs
CREATE TABLE all_sales AS
SELECT
    InvoiceNo,
    StockCode,
    Quantity,
    UnitPrice,
    (Quantity * UnitPrice) AS Revenue, -- Feature Engineering
    InvoiceDate,
    CASE
        WHEN CustomerID IS NULL OR CustomerID = '' THEN 'Guest'
        ELSE CAST(CustomerID AS CHAR)
    END AS CustomerID, -- Handling Missing Values
    Country
FROM retail_data
WHERE Quantity > 0; -- Delete Return/Cancellations
```


Seasonal Trends & Market Performance

Revenue by Country

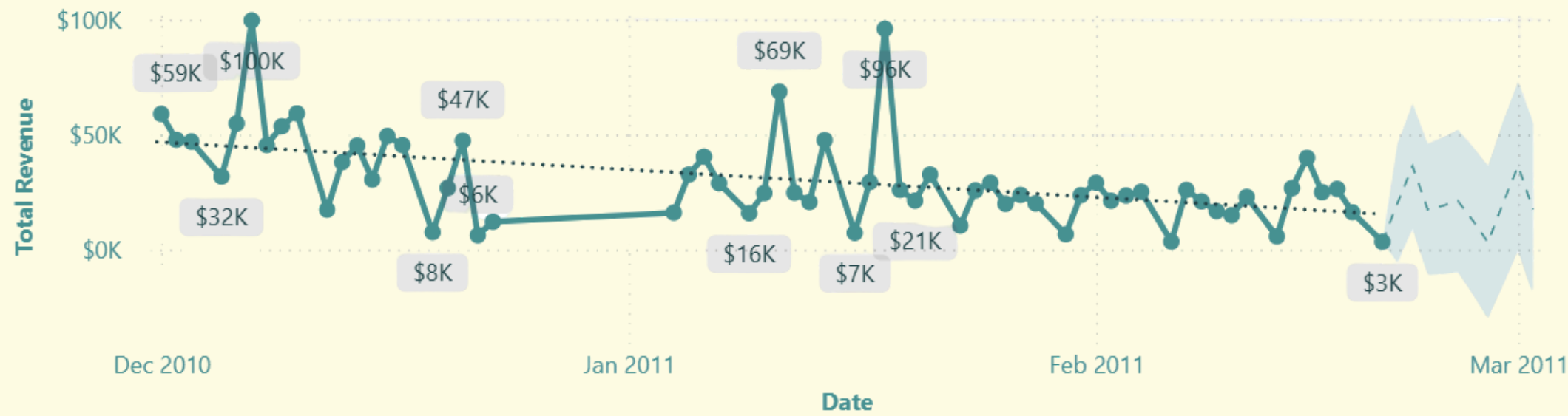


Market Dominance: The UK stands as the primary revenue hub, contributing significantly to the total \$1.86M revenue.

Growth Potential: International markets like EIRE (Ireland) and Germany show high transaction quality despite lower volume.

Key Finding: December revenue (\$823K) represents a 137% surge compared to the February low, indicating a strong holiday-driven business model with significant Q1 seasonality risks.

Revenue Trend



Analytics Methodology: RFM Segmentation Framework

- **METHODOLOGY:**
 - **Recency (R):** Evaluated days since the last transaction. Lower values = Higher engagement.
 - **Frequency (F):** Measured total distinct InvoiceNo per customer to identify loyalty.
 - **Monetary (M):** Calculated total lifetime spend per customer.
 - **Scoring System:** Applied Statistical Quintiles (NTILE-5) to rank customers from 1–5 across all metrics.

```
-- RFM Scoring using Statistical Quintiles
WITH rfm_score AS (
    SELECT
        CustomerID,
        NTILE(5) OVER(ORDER BY Recency) AS R,
        NTILE(5) OVER(ORDER BY Frequency DESC) AS F,
        Total_Revenue AS M
    FROM rfm_values)
SELECT * FROM rfm_score;
```

Customer Segmentation Results

Customer Distribution by Segment

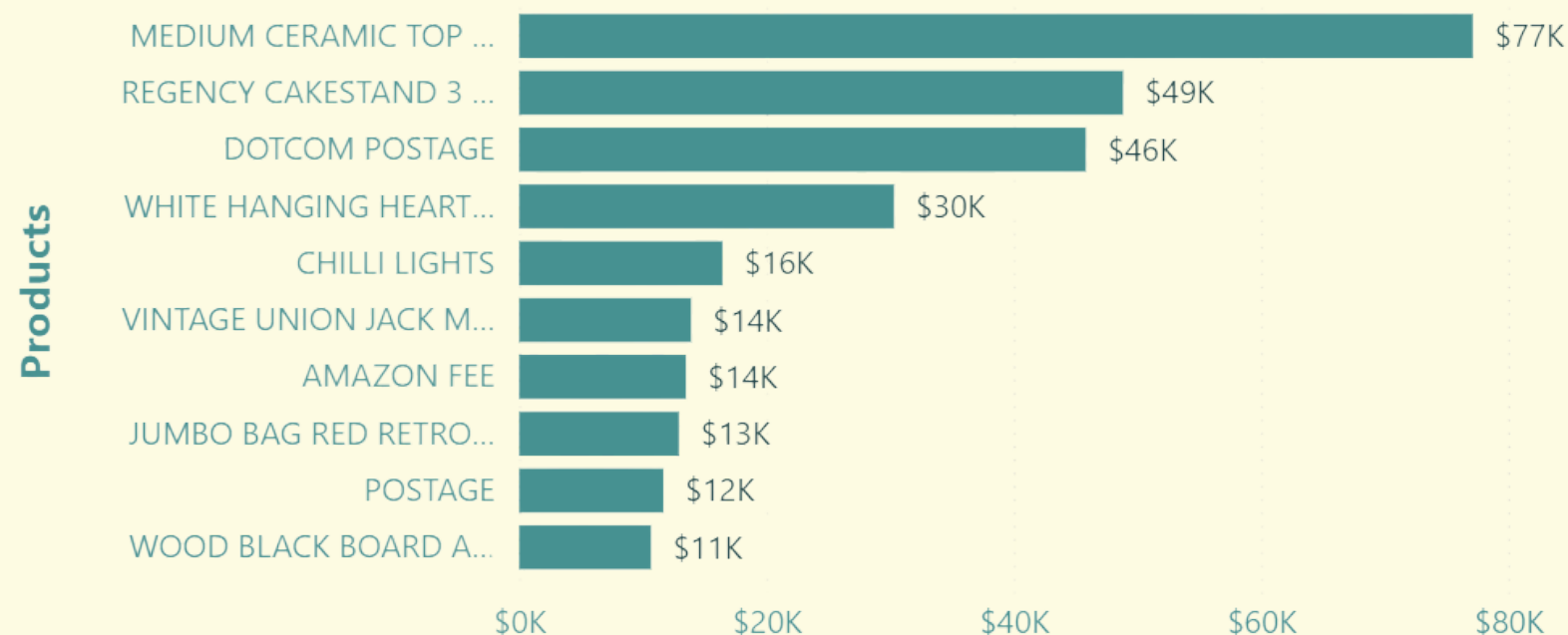


- **Champions & Loyalists:** Identified the core user base driving the majority of recurring revenue. These segments exhibit high frequency and recent activity.
- **At Risk / Need Attention:** A significant volume of customers has not purchased in 90+ days. This represents a primary target for re-engagement.
- **Potential Loyalists:** High-spending new customers who require "nurturing" to transition into the Loyalist category.

The "Champions" and "Loyal Customers" segments represent the highest LTV (Lifetime Value), while the "At Risk" group indicates a potential revenue leakage that requires immediate intervention.

Top 10 Products by Revenue

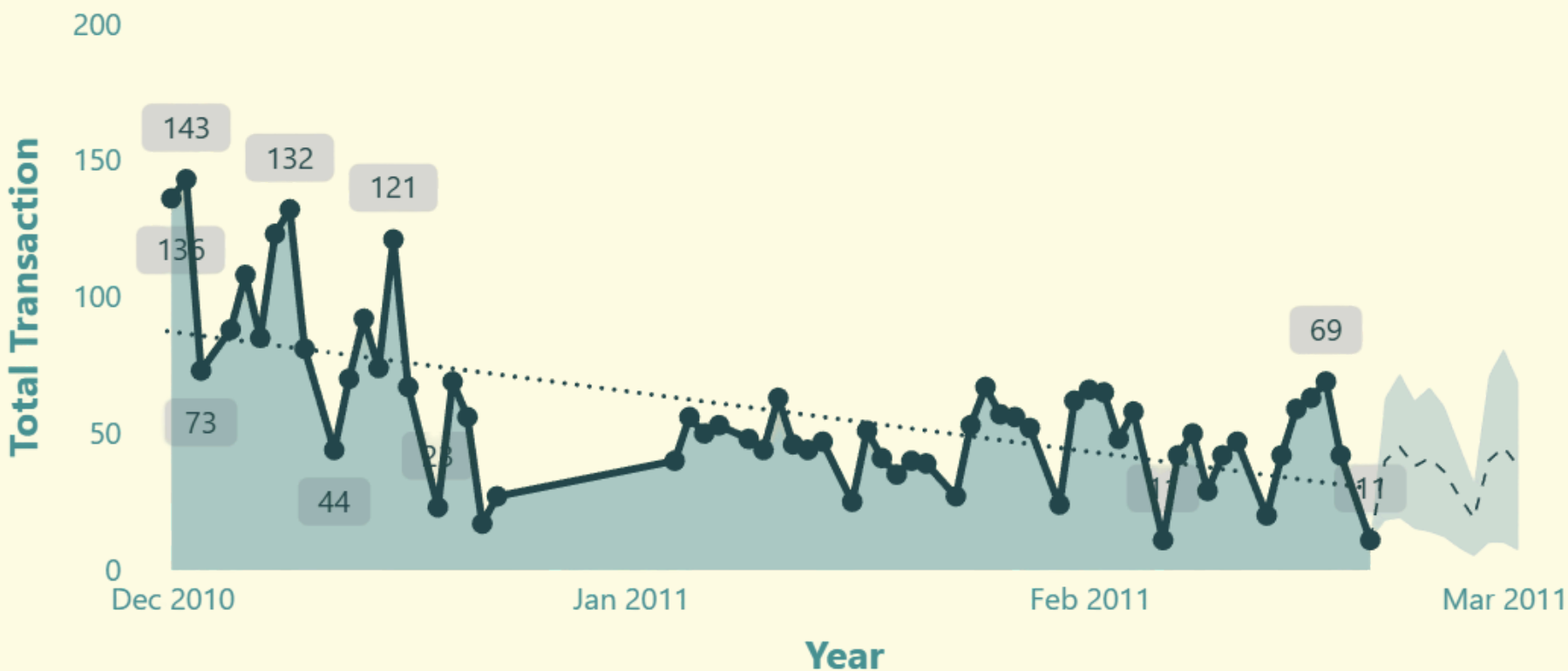
Top 10 Products by Revenue



Diversifying high-revenue products and optimizing postage/shipping logic could increase overall profit margins by ~5-10% without increasing sales volume.

- **Top Performers:** High-demand items like "Regency Cakestand" and "White Hanging Heart" are consistent volume drivers.
- **Operational Overhead:** Analysis of "DOTCOM POSTAGE" and "AMAZON FEE" reveals significant logistics costs impacting net margins.
- **Inventory Velocity:** Transaction patterns show high volatility, necessitating a dynamic stock replenishment strategy.

Overtime Transaction Patterns



HIGH-VALUE RETENTION (CHAMPIONS & LOYALISTS)

- **ACTION:** LAUNCH AN EXCLUSIVE VIP LOYALTY PROGRAM.
- **GOAL:** OFFER EARLY ACCESS TO HOLIDAY COLLECTIONS AND PERSONALIZED REWARDS TO MAXIMIZE CUSTOMER LIFETIME VALUE (LTV).

CHURN MITIGATION (AT RISK & ABOUT TO SLEEP)

- **ACTION:** DEPLOY AUTOMATED WIN-BACK CAMPAIGNS.
- **GOAL:** TRIGGER PERSONALIZED EMAIL DISCOUNTS (E.G., 15% OFF) FOR CUSTOMERS WHO HAVEN'T PURCHASED IN 90+ DAYS TO PREVENT PERMANENT CHURN.

Strategic Recommendations

OPERATIONAL MARGIN REVIEW (LOGISTICS)

- **ACTION:** OPTIMIZE POSTAGE & AMAZON FEES.
- **GOAL:** REVIEW LOGISTICS CONTRACTS AND IMPLEMENT A "FREE SHIPPING THRESHOLD" (E.G., SPEND > \$50) TO OFFSET HIGH SHIPPING OVERHEADS.

INVENTORY CROSS-SELLING

- **ACTION:** STRATEGIC PRODUCT BUNDLING.
- **GOAL:** PAIR "STAR PRODUCTS" (REGENCY CAKESTANDS) WITH LOWER-VELOCITY DECOR ITEMS TO CLEAR SLOW-MOVING STOCK AND INCREASE AVERAGE ORDER VALUE (AOV).

Bridging Theory & Data

BY INTEGRATING DEVELOPMENT ECONOMICS INTUITION WITH TECHNICAL DATA ENGINEERING, THIS PROJECT SUCCESSFULLY TRANSFORMED 500K+ RAW TRANSACTIONS INTO A STRUCTURED GROWTH ROADMAP. THE COMBINATION OF SQL-DRIVEN ETL AND POWER BI STORYTELLING PROVIDES THE CLARITY NEEDED FOR DATA-BACKED EXECUTIVE DECISION-MAKING.

LET'S CONNECT & WORK TOGETHER!

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